Ttile: A localized particle filter for geophysical data assimilation

Link: http://arxiv.org/abs/2507.07103v1

Abstract: Particle filters are computational techniques for estimating the state of dynamical systems by integrating observational data with model predictions . This work introduces a class of Localized Particle Filters (LPFs) that exploit spatial localization to reduce computational costs .

Ttile: Non-Gaussian Phase Transition and Cascade of Instabilities in the Dissipative Quantum Rabi Model

Link: http://arxiv.org/abs/2507.07092v1

Abstract: The open quantum Rabi model describes a two-level system coupled to a harmonic oscillator . A Gaussian phase transition for the nonequilibrium steady states has been predicted when the bosonic mode is soft and subject to damping .

Ttile: Investigating the CREDIT history of supernova remnants as cosmic-ray sources

Link: http://arxiv.org/abs/2507.07088v1

Abstract: Supernova remnants (SNRs) have long been suspected to be the primary sources of Galactic cosmic rays . Over the past decades, great strides have been made in modelling of particle acceleration, magnetic field amplification, and escape from SNRs. Yet there is no evidence for any individual object contributing to the locally observed flux .

Ttile: An AI Approach for Learning the Spectrum of the Laplace-Beltrami Operator

Link: http://arxiv.org/abs/2507.07073v1

Abstract: the best established method for its estimation is based on the Finite Element Method (FEM) and computes the top k LB eigenvalues with a complexity of O(Nk) This can render the FEM method inefficient when repeatedly applied to databases of CAD mechanistic data .

Ttile: How to Bridge the Sim-to-Real Gap in Digital Twin-Aided Telecommunication Networks

Link: http://arxiv.org/abs/2507.07067v1

Abstract: training effective artificial intelligence models for telecommunications is challenging due to the scarcity of deployment-specific data . Real data collection is expensive, and available datasets often fail to capture the unique operational conditions and contextual variability of the network environment .